equipped with sophisticated control, communication, propulsion, and payload technologies that enabled enhanced stability, longer endurance, and semi-autonomous flight capabilities. Each component within these systems played a vital role in ensuring operational precision, mission versatility, and improved performance under varied environmental and tactical conditions.

- *Airframe*: The airframe of second-generation drones was primarily constructed using lightweight composite materials such as carbon fiber, fiberglass, and aluminum alloys to maximize structural strength while minimizing weight. These aerodynamic designs were optimized for endurance, agility, and reduced drag, ensuring smoother flight dynamics and improved fuel efficiency. The shift from wooden or fabric structures of first-generation UAVs to composites marked a turning point in drone durability and performance.
- *Propulsion system*: Propulsion systems included piston engines, turbojets, or electric motors, depending on mission requirements. These systems provided higher power-to-weight ratios and allowed for extended flight times. Improved engine efficiency and reliability supported longer missions with greater payload capacity, enabling drones to operate at higher altitudes and speeds for reconnaissance, surveillance, and mapping operations.
- *Flight controller (FCU)*: The flight control unit (FCU) functioned as the UAV's central processing system, or "brain." It interpreted commands from the operator or autopilot system and coordinated control surface movements to maintain stability and direction. By integrating data from various sensors such as gyroscopes, accelerometers, and barometers, the FCU enabled semi-autonomous flight, route correction, and altitude control, significantly reducing pilot workload.